



BUCKET SORT 5

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Given N floating-point numbers in the range $[0, 1)$, sort them using the Bucket Sort algorithm.

Input Format

N a_1 a_2 a_3 ... a_N

Constraints

$1 \leq N \leq 100000$ $0 \leq a_i < 1$

Output Format

Print the sorted array rounded to 2 decimal places.

Sample Input 0

```
6
0.90 0.80 0.70 0.60 0.50 0.40
```

Sample Output 0

```
0.40 0.50 0.60 0.70 0.80 0.90
```

Sample Input 1

```
6
10 20 40 50 30 60
```

Sample Output 1

```
10 20 30 40 50 60
```

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Language

Python 3



```
1 import sys
2
3 def bucket_sort(arr):
4     if not arr:
5         return arr
6
7     n = len(arr)
8     min_val, max_val = min(arr), max(arr)
9
10    if min_val == max_val:
11        return arr
```

```
12
13     buckets = [[] for _ in range(n)]
14     val_range = max_val - min_val
15
16     for x in arr:
17
18         idx = int((x - min_val) / val_range * (n - 1))
19         buckets[idx].append(x)
20
21     sorted_arr = []
22     for bucket in buckets:
23         sorted_arr.extend(sorted(bucket))
24
25     return sorted_arr
26
27 def solve():
28     input_data = sys.stdin.read().split()
29     if not input_data:
30         return
```

Line: 46 Col: 1

 Upload Code as File☐ Test against custom input

Run Code

Submit Code

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

 Sample Test case 0 Sample Test case 1

Input (stdin)

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```
1 | 6
2 | 0.90 0.80 0.70 0.60 0.50 0.40
```

Your Output (stdout)

```
1 | 0.40 0.50 0.60 0.70 0.80 0.90
```

Expected Output

Download

```
1 | 0.40 0.50 0.60 0.70 0.80 0.90
```